Haversine Formula: Step-by-Step Computation
Edinburgh (55.95°N, 3.19°W) → London (51.51°N, 0.13°W)

Step 1: $\Delta\phi$ (delta-phi) $= \phi_2 - \phi_1 = 51.51° - 55.95° = -4.44°\ = -0.0775$ rad

Step 2: $\Delta\lambda$ (delta-lambda) $= \lambda_2 - \lambda_1 = -0.13° - (-3.19°) = 3.06°\ = 0.0534$ rad

Step 3: $a = \sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2)$

Step 4: $a = \sin^2(-0.0388) + \cos(0.977) \cdot \cos(0.899) \cdot \sin^2(0.0267)$

Step 5: $a = 0.001505 + 0.5553 \times 0.6248 \times 0.000712 = 0.001505 + 0.000247 = 0.001752$

Step 6: c (central angle) $= 2\,\mathrm{atan2}(\sqrt{a},\ \sqrt{1-a}) = 2\,\mathrm{atan2}(0.04181,\ 0.9991) = 0.08363$ rad

Step 7: $d = R \times c = 6371 \times 0.08363 \approx \mathbf{532.7}$ km